

```

Program Fixed_point
  Implicit none
  Real(8)::p0
  Integer::max_iter
  Write(*,*)"Enter the p0"
  Read(*,*)p0
  Write(*,*)"Enter the max_iter"
  Read(*,*)max_iter
  Call solve_fixed_point(p0,max_iter,"g1")
  call solve_fixed_point(p0,max_iter,"g2")
  call solve_fixed_point(p0,max_iter,"g3")

End Program fixed_point

Subroutine solve_fixed_point(p0,max_iter,func_name)
  Real(8),Intent(IN)::p0
  Integer,Intent(IN)::max_iter
  character(*),Intent(IN)::func_name
  real(8)::pn,pn1,error
  real(8)::g1,g2,g3
  integer::i
  pn=p0
  error=1.0
  i=0
  Write(*,*)"Solving using",func_name
  Write(*,*)"Iter  p(n-1)  pn  Error"
  Do while(error>1.0d-6 .and. i<max_iter)
    if(func_name== "g1")then
      pn1=g1(pn)

    else if(func_name=="g2")then
      pn1=g2(pn)
    else if(func_name=="g3")then
      pn1=g3(pn)
    else
      Write(*,*)"Invalid"
      return
    end if

    error=Abs(pn1-pn)
    Write(*,'(I10,3F12.6)')i,pn1,pn,error
    pn=pn1
    i=i+1

  End Do
  If(error<=1.0d-6)then
    Write(*,*)"Congverged Solution pn1=",pn1
  else
    Write(*,*)"Do not Converge"
  End If

End Subroutine

real(8) function g1(x)
real(8), intent(in) :: x
g1 = (x + 10.0)**0.25
end function g1

real(8) function g2(x)
real(8), intent(in) :: x

```

```
if (x==0)then
write(*,*)"Warning: g2(x) is undefined at x = 0. Returning a default value."
else
g2 = sqrt(x + 10.0) / x
end if
end function g2

real(8) function g3(x)
real(8), intent(in) :: x
if (x**3 == 1.0) then
write(*,*)"Warning: g3(x) is undefined at x = 1. Returning a default value."
else
g3 = 10.0 / (x**3 - 1.0)
end if
end function g3
```